

AOTS2Action: Uncertainty-Aware Big Data Analytics for Humanitarian Cyclone Risk

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Abstract—Ensemble tropical cyclone forecasts represent multiple plausible future trajectories, but downstream humanitarian analyses typically reduce this uncertainty to a deterministic track or a single uncertainty region before estimating exposure. Such early aggregation discards information about how strongly the forecast supports impact at each location, and can therefore change which populations and regions are prioritized for preparedness. We present Ahead-of-the-Storm-to-Action (AOTS2Action), an uncertainty-preserving geospatial analytics framework that propagates tropical cyclone ensemble uncertainty through heterogeneous data describing population, vulnerability, critical infrastructure, and administrative regions. The framework retains individual ensemble members across forecast horizons; supports fixed-radius, percentile-based, and covariance-ellipse uncertainty representations within a common pipeline; and computes probability-weighted population and vulnerability-weighted exposure that distinguishes locations strongly supported by the ensemble from locations that fall within a broad uncertainty envelope but have limited ensemble support. These spatial estimates are aggregated into uncertainty-aware regional priority rankings. We evaluate AOTS2Action retrospectively by matching ensemble forecasts for [N storms / N forecast cycles] against best-track observations, measuring empirical P50 and P90 coverage, calibration error, uncertainty-region sharpness, and great-circle track error across forecast horizons, together with exposure-estimation error relative to realized-track exposure and regional-ranking quality via Spearman correlation, nDCG@K, and Recall@K. The uncertainty-aware analysis achieves [X%] empirical P90 coverage at [H]-h lead time while reducing uncertainty-region area by [Y%] relative to [NAMED BASELINE]. Probability-weighted exposure reduces exposure-estimation error by [Z%] relative to exposure computed from the [deterministic baseline: ensemble-mean track / control member], and regional prioritization reaches nDCG@10 = [R] at [H]-h lead time. Scalability experiments characterize runtime, memory, and throughput as ensemble size and geospatial data volume increase. These results indicate that preserving forecast uncertainty throughout the geospatial analytics pipeline provides a better-calibrated and more discriminative basis for humanitarian exposure estimation and regional prioritization than collapsing ensemble forecasts before impact analysis.

Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

Tropical cyclones (TCs) are high-impact natural hazards whose societal consequences depend not only on cyclone intensity and trajectory, but also on the spatial distribution of exposed populations, socioeconomic vulnerability, and critical

infrastructure. Effective preparedness therefore requires more than predicting where a cyclone may travel: decision makers must understand the range of plausible future trajectories, which communities may be affected under those trajectories, and how forecast uncertainty changes across planning horizons.

Recent advances in artificial intelligence have substantially improved data-driven weather forecasting. Pangu-Weather [1] and GraphCast [2] demonstrated that learned global forecasting systems can achieve highly competitive medium-range prediction while substantially reducing inference cost. NeuralGCM [3] combined differentiable physical dynamics with machine learning and demonstrated both deterministic and ensemble forecasting capabilities. More recently, GenCast [4] formulated global weather forecasting explicitly as a probabilistic ensemble-generation problem, while Aurora [5] demonstrated the potential of foundation models across multiple Earth-system prediction tasks, including tropical cyclone track forecasting. Collectively, these developments make large ensembles of high-resolution forecast trajectories increasingly available for uncertainty-aware downstream applications. While the framework presented here is forecast-source-agnostic, this growing availability of ensemble output makes uncertainty-preserving downstream analytics increasingly consequential.

For tropical cyclones specifically, accurately representing *where the cyclone may go* remains distinct from translating that uncertainty into *who and what may be affected*. Recent studies have therefore focused on improving track-uncertainty representation. Yoo and Ho [6] developed interpretable probability ellipses from multiple operational ensemble prediction systems, showing the value of representing anisotropic along-track and cross-track uncertainty rather than relying exclusively on circular regions. Meng and Song [7] introduced conformal prediction for TC track uncertainty and constructed trajectory forecast ellipses with statistically motivated uncertainty regions. Fernandez *et al.* [8] proposed a probabilistic neural model that predicts dynamic bivariate track-error distributions and supports calibrated uncertainty estimation and landfall probabilities. These studies demonstrate the importance of probabilistic track forecasting, but their primary objective is characterization of cyclone-position uncertainty

rather than systematic propagation of that uncertainty into population exposure, vulnerability, infrastructure, and regional humanitarian priorities.

At the same time, big-data research has increasingly connected environmental hazards with downstream infrastructure and geospatial information. Lee *et al.* [9] integrated hurricane advisories, power-system information, and outage data within geographic graph-based learning models for county-level outage prediction during extreme events, and Adams *et al.* [10] incorporated hierarchical spatial context to infer missing building characteristics across millions of structures. These studies illustrate the value of combining heterogeneous spatial datasets with machine learning for climate-risk and disaster applications. An important gap nevertheless remains between probabilistic cyclone forecasting and downstream humanitarian analytics: in existing operational and research workflows, forecast uncertainty is typically summarized—as a mean track, a fixed cone, or a small set of discrete scenarios—*before* exposure analysis, rather than propagated through the complete decision-support pipeline.

Closing this gap is a characteristic big-data problem. A single cyclone forecast may contain many ensemble members across multiple initialization times and forecast horizons, while humanitarian analysis requires their integration with population rasters, vulnerability indicators, administrative boundaries, and infrastructure records at different spatial resolutions and levels of completeness. The challenge therefore involves not only data *volume* and *variety*, but also *veracity*, because forecast uncertainty and source-data quality directly affect downstream estimates, and *velocity*, because the analysis must be repeated as new forecast cycles become available. Collapsing an ensemble prematurely to a single mean trajectory, or treating all locations inside an uncertainty region as equally likely, discards information that can change humanitarian prioritization.

We present **Ahead-of-the-Storm-to-Action (AOTS2Action)**, an uncertainty-preserving big-data analytics framework that transforms tropical cyclone ensemble forecasts into multi-horizon humanitarian risk information. Rather than treating forecast uncertainty as an endpoint of the forecasting process, AOTS2Action retains individual ensemble members throughout the analytics pipeline and propagates their spatial uncertainty into population exposure and vulnerability-aware risk estimation. The framework integrates ensemble trajectories with heterogeneous geospatial datasets describing population, vulnerability, administrative regions, and critical infrastructure, and produces multi-resolution outputs ranging from forecast-level uncertainty estimates to region-level humanitarian priorities.

AOTS2Action supports several representations of forecast uncertainty within a single pipeline, including deterministic ensemble means, fixed-radius regions, percentile-based uncertainty cones, and covariance-based probability ellipses. More importantly, it derives spatial exposure probabilities directly from ensemble support and uses them to compute expected population exposure and vulnerability-weighted exposure. This

formulation distinguishes locations strongly supported by the ensemble from locations that fall within a broad uncertainty envelope but have limited ensemble support—a distinction that deterministic tracks and uniform uncertainty regions cannot express. Regional aggregation then converts these spatial estimates into ranked administrative-area priorities suitable for downstream decision-support applications.

We evaluate the framework retrospectively along four complementary dimensions. Our evaluation uses **[ENSEMBLE SOURCE, e.g., GenCast / ECMWF EPS / operational multi-model]** ensemble forecasts for **[N]** tropical cyclones in the **[BASIN(S)]** basin(s) during **[YEARS]**, matched against best-track observations from **[IBTrACS / HURDAT2 / JTWC]**. First, we assess uncertainty calibration by measuring empirical P50 and P90 coverage—the frequency with which observed cyclone positions fall within the nominal 50% and 90% uncertainty regions—together with calibration error, uncertainty-region area, and great-circle track error across forecast horizons. Second, we compare deterministic, P90-region, and probability-weighted ensemble approaches for population and vulnerability-weighted exposure estimation against realized-track exposure. Third, we evaluate regional prioritization against retrospective rankings derived from realized cyclone trajectories using Spearman correlation, $nDCG@K$, and $Recall@K$. Finally, because repeated ensemble-geospatial integration is computationally intensive, we characterize system scalability as ensemble size and spatial data volume increase.

The main contributions of this work are:

- **Uncertainty-preserving analytics with probability-weighted exposure.** We develop an end-to-end framework that retains full tropical cyclone ensembles throughout the analytics pipeline and derives spatial exposure probabilities directly from ensemble support, yielding expected population exposure and vulnerability-weighted risk estimates that distinguish strongly supported locations from weakly supported regions of the uncertainty envelope.
- **Uncertainty-aware integration and regional prioritization at scale.** We integrate ensemble forecasts with heterogeneous population, vulnerability, infrastructure, and administrative data under multiple interchangeable uncertainty representations, and convert the resulting spatial risk estimates into region-level humanitarian priority rankings while preserving data provenance.
- **A retrospective evaluation protocol.** We introduce a multi-dimensional protocol that jointly assesses uncertainty calibration and sharpness, exposure-estimation fidelity against realized trajectories, regional-ranking quality, and computational scalability, enabling systematic and reusable comparison between deterministic and uncertainty-aware approaches.

Overall, AOTS2Action bridges probabilistic tropical cyclone forecasting and humanitarian risk analytics by treating uncertainty as information to be propagated through the data

pipeline rather than removed before impact assessment. The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the AOTS2Action framework and its uncertainty representations. Section IV describes the retrospective evaluation methodology, Section V reports results, and Section VII concludes.

II. RELATED WORK

We organize related work into four areas: probabilistic and data-driven weather forecasting, tropical cyclone track-uncertainty representation, impact-based forecasting and anticipatory humanitarian action, and large-scale geospatial analytics for disaster risk. AOTS2Action draws on all four but occupies a position none of them individually covers: the propagation of member-level ensemble uncertainty through exposure estimation and regional prioritization, with joint evaluation of calibration, exposure fidelity, ranking quality, and scalability.

A. Probabilistic and Data-Driven Weather Forecasting

Operational ensemble prediction systems have long provided the primary source of forecast uncertainty information for tropical cyclones, and multi-decadal intercomparison shows a steady increase in the ability of global models to predict TC positions [11], with multi-model ensembles further improving track probability forecasts [12]. Machine-learned forecasting systems have recently altered the cost structure of ensemble generation. Pangu-Weather [1] and GraphCast [2] matched or exceeded operational deterministic skill at a small fraction of the inference cost; NeuralGCM [3] combined differentiable dynamics with learned components and demonstrated ensemble capability; and GenCast [4] generated calibrated probabilistic forecasts by design. Aurora [5] extended foundation-model approaches to multiple Earth-system tasks, including TC track prediction. Because inference is inexpensive, such models make very large ensembles practical, which increases—rather than reduces—the importance of downstream analytics that can consume full ensembles without early aggregation. AOTS2Action is forecast-source-agnostic and treats these systems as upstream producers: our contribution begins where theirs ends, at the point where an ensemble of trajectories must be converted into humanitarian information.

B. Tropical Cyclone Track-Uncertainty Representation

A substantial body of work addresses how TC positional uncertainty should be represented and communicated. The National Hurricane Center’s cone of uncertainty, constructed from historical track-error quantiles, remains the most widely disseminated representation, and its frequent misinterpretation by the public and decision makers is well documented [13]. Wind-speed probability products [14] moved beyond the cone by expressing exceedance probabilities at fixed locations, derived from Monte Carlo sampling of track, intensity, and wind-structure error distributions. More recently, Yoo and

Ho [6] constructed interpretable probability ellipses from multiple operational ensembles, capturing anisotropic along-track and cross-track error; Meng and Song [7] applied conformal prediction to obtain distribution-free track uncertainty regions with finite-sample coverage guarantees; and Fernandez *et al.* [8] predicted dynamic bivariate track-error distributions supporting calibrated landfall probabilities.

These methods improve the fidelity and calibration of *position* uncertainty, and AOTS2Action deliberately incorporates their representational vocabulary: fixed-radius regions, percentile-based cones, and covariance ellipses are all supported as interchangeable uncertainty representations within our pipeline (Section III). The distinction is one of scope. Prior work in this line terminates at the uncertainty region itself, whereas our concern is what happens *after*: whether exposure, vulnerability, and regional priority estimates computed from a representation inherit its calibration, and how much information is lost when the representation replaces the underlying ensemble. We are not aware of prior work on tropical cyclones that jointly evaluates track-uncertainty calibration and the downstream population-exposure and prioritization estimates it induces within a single retrospective protocol.

C. Impact-Based Forecasting and Anticipatory Action

The humanitarian community has increasingly shifted from hazard-centric warning toward impact-based forecasting and forecast-based (anticipatory) action, in which pre-agreed measures are triggered when forecasts cross activation thresholds. The WMO guidelines on impact-based forecasting [15] formalize the shift from *what the weather will be* to *what the weather will do*. Operational systems embody this shift in different ways. GDACS [16] issues automated multi-hazard alerts by combining physical hazard parameters with demographic and socioeconomic data to estimate expected impact; forecast-based financing [17] triggers pre-agreed humanitarian funding when forecast probabilities exceed thresholds chosen to balance the costs of acting in vain against acting too late; and the framework name used here extends the Ahead-of-the-Storm (AOTS) initiative [18], which produces child-focused impact assessments ahead of tropical cyclones using discrete forecast scenarios (expected, best, and worst case). Academic work in this space includes county-level infrastructure-impact prediction such as the hurricane–power-outage models of Lee *et al.* [9].

In each of these designs, the ensemble or probabilistic forecast is reduced at a specific point before impact analysis: to an alert-level category in GDACS, to a binary threshold crossing in forecast-based financing triggers, and to a small discrete scenario set in AOTS. Such reductions are operationally motivated and often appropriate for their decision contexts, but they share a consequence: the resulting exposure estimates carry no notion of differential forecast support across locations, and their calibration against realized outcomes is rarely quantified. AOTS2Action complements these systems by retaining member-level ensemble support through exposure estimation and regional ranking, deriving exposure *proba-*

bilities rather than scenario exposures, and evaluating both calibration and ranking quality against realized trajectories.

D. Large-Scale Geospatial Analytics for Disaster Risk

Translating hazard estimates into impact estimates requires integrating heterogeneous geospatial data at scale, a problem the big-data community has engaged from several directions. Gridded population products such as WorldPop [19] and the Global Human Settlement Layer population grids [20] provide the exposure substrate; composite indicators such as the INFORM risk index [21] summarize hazard, vulnerability, and coping capacity at country level; and open platforms such as CLIMADA [22] integrate hazard, exposure, and vulnerability in an event-based probabilistic framework, including synthetic TC track generation for long-run risk assessment. Recent work at this venue illustrates the analytics challenges of such integration: Lee *et al.* [9] fused hurricane advisories with power-system and outage records in geographic graph-learning models, and Adams *et al.* [10] used hierarchical spatial context to infer missing building attributes across millions of structures, directly confronting the veracity and completeness problems that characterize large geospatial corpora.

Catastrophe-modeling platforms such as CLIMADA are the closest in spirit to our exposure computation, but they primarily target climatological risk—expected impacts over large synthetic event sets—rather than forecast-time analytics for a specific evolving storm, and they are not designed to preserve and evaluate the calibration of an operational ensemble across forecast horizons. AOTS2Action addresses the forecast-time setting: repeated ensemble–geospatial integration under velocity constraints, with provenance tracking across sources of differing resolution and completeness, and with scalability characterized explicitly as ensemble size and data volume grow.

E. Positioning

In summary, prior work provides high-quality ensemble forecasts (§2.1), well-calibrated representations of positional uncertainty (§2.2), operational impact-based products that reduce forecast uncertainty at a defined point before impact analysis (§2.3), and large-scale geospatial integration machinery (§2.4). AOTS2Action connects these threads: it consumes full ensembles, supports multiple uncertainty representations under a common interface, propagates member-level support into probability-weighted exposure and vulnerability estimates, aggregates them into regional priorities, and evaluates the entire chain retrospectively—calibration, exposure fidelity, ranking quality, and scalability—against realized cyclone trajectories. We are not aware of a prior framework that combines this end-to-end uncertainty-preserving formulation with a joint evaluation of all four dimensions.

III. THE AOTS2ACTION FRAMEWORK

AOTS2Action converts an ensemble of tropical cyclone track forecasts and a harmonized collection of geospatial layers into uncertainty-aware exposure estimates and regional

humanitarian priorities. Figure ?? summarizes the pipeline: ensemble ingestion and normalization, geospatial harmonization, uncertainty representation, probability-weighted exposure estimation, and regional aggregation. The design principle throughout is that individual ensemble members remain first-class objects until the final aggregation step; uncertainty representations are derived *views* of the ensemble used for comparison and communication, not replacements for it.

A. Problem Setting and Notation

Let $\mathcal{E} = \{T_1, \dots, T_M\}$ denote an ensemble of M forecast trajectories issued at initialization time t_0 . Each member T_m is a sequence of forecast positions $\mathbf{p}_m(h) = (\phi_m(h), \lambda_m(h))$ in geographic coordinates (latitude, longitude) at lead times (horizons) $h \in \mathcal{H} = \{h_1, \dots, h_L\}$. Members need not span all horizons; we write $\mathcal{M}_h$ for the set of members with a position at horizon h and $M_h = |\mathcal{M}_h|$. Distances are great-circle distances $d(\cdot, \cdot)$.

Geospatial context is provided as a harmonized analysis representation $\mathcal{X}$ whose elements x carry geographic coordinates, population $P(x) \geq 0$, a vulnerability indicator $V(x) \in [0, 1]$, administrative-region membership $R(x) \in \{R_1, \dots, R_J\}$, and an infrastructure count $I(x) \geq 0$. All layers carry provenance metadata (source, version, retrieval date, native resolution, and any transformation applied), which is propagated to every derived product so that downstream estimates remain auditable.

B. Ensemble Ingestion and Geospatial Harmonization

Forecast ensembles arrive in heterogeneous formats and conventions (**[e.g., ATCF a-decks, NetCDF, provider-specific CSV]**). Ingestion normalizes coordinate conventions and horizon grids into the trajectory representation of Section III-A; members without a position at a given horizon are simply absent from $\mathcal{M}_h$, and all ensemble statistics are computed over the available members at each horizon.

Geospatial sources are preprocessed into a common analysis representation $\mathcal{X}$ before forecast evaluation. Each analysis element contains geographic coordinates, population, vulnerability, administrative-region membership, and infrastructure attributes. Source-specific preprocessing, including reprojection, spatial aggregation, and administrative joins, is performed once and recorded through dataset provenance metadata. Forecast evaluation therefore operates on a fixed harmonized representation while ensemble-dependent computations are repeated for each forecast cycle, addressing the velocity requirement identified in Section I.

C. Uncertainty Representations

For comparison and communication, AOTS2Action derives four representations from the ensemble at each horizon h .

a) *Deterministic ensemble mean.*: At horizon h , the representative location is obtained by the component-wise mean latitude and longitude of the available ensemble members,

$$\bar{\mathbf{p}}(h) = \left(\frac{1}{M_h} \sum_{m \in \mathcal{M}_h} \phi_m(h), \frac{1}{M_h} \sum_{m \in \mathcal{M}_h} \lambda_m(h) \right). \quad (1)$$

When the ensemble is multimodal (e.g., recurvature-versus-landfall splits), the mean can fall in a region supported by few or no members; this is one motivation for the ensemble-support formulation of Section III-D.

b) *Fixed-radius region.*: The disc $B(\bar{\mathbf{p}}(h), r_0)$ of constant radius $r_0 = [\text{VALUE}]$ km around the mean track. This is a non-probabilistic geometric baseline, mirroring simple buffer-based operational practice; it makes no nominal coverage claim, and its empirical coverage is measured retrospectively in Section V.

c) *Percentile-based cone.*: For a nominal level $\alpha \in \{0.5, 0.9\}$, the cone radius at horizon h is the empirical α -quantile of member-mean distances plus a pre-specified base buffer b ,

$$r_\alpha(h) = Q_\alpha(\{d(\mathbf{p}_m(h), \bar{\mathbf{p}}(h))\}_{m \in \mathcal{M}_h}) + b, \quad (2)$$

yielding horizon-dependent circular regions $C_\alpha(h) = B(\bar{\mathbf{p}}(h), r_\alpha(h))$ analogous to the operational cone but derived from the live ensemble rather than climatological error statistics.

d) *Covariance ellipse.*: Member positions at horizon h are projected into a local equirectangular coordinate system centered on the ensemble mean, and their 2×2 sample covariance $\Sigma(h)$ defines elliptical regions

$$E_\alpha(h) = \{\mathbf{x} : (\mathbf{x} - \bar{\mathbf{p}}(h))^\top \Sigma(h)^{-1} (\mathbf{x} - \bar{\mathbf{p}}(h)) \leq \chi_2^2(\alpha)\}, \quad (3)$$

where $\chi_2^2(\alpha)$ is the α -quantile of the chi-squared distribution with two degrees of freedom. Ellipses capture the anisotropic along-track/cross-track structure emphasized in prior work [6], at the cost of a bivariate-normal assumption that the calibration evaluation of Section IV tests empirically rather than presumes.

The probabilistic representations (cone and ellipse) define binary region memberships that treat all covered locations as equally likely — precisely the simplification that the ensemble-support formulation removes.

D. Ensemble-Derived Exposure Probabilities

The central object in AOTS2Action is the *exposure probability* of a location: the fraction of ensemble support under which the cyclone passes close enough to x to plausibly affect it. With impact radius $r = [\text{VALUE}]$ km, the per-horizon exposure probability is

$$q_h(x) = \frac{1}{M_h} \sum_{m \in \mathcal{M}_h} \mathbb{I}[d(x, \mathbf{p}_m(h)) \leq r]. \quad (4)$$

Equation (4) answers the instantaneous question: what may be affected *at* lead time h .

Humanitarian planning also requires the cumulative question: what may be affected at *any point* through lead time H . Let $\mathcal{M}_{\leq H}$ denote the set of members with positions at every horizon $h \in \mathcal{H}$ with $h \leq H$. The through-horizon exposure probability is

$$q_{\leq H}(x) = \frac{1}{|\mathcal{M}_{\leq H}|} \sum_{m \in \mathcal{M}_{\leq H}} \mathbb{I}\left[\min_{\substack{h \in \mathcal{H} \\ h \leq H}} d(x, \mathbf{p}_m(h)) \leq r\right]. \quad (5)$$

Equation (5) is computed *per member* before averaging. This preserves trajectory coherence: a location is exposed under member m if that member's track ever approaches it, which is not equivalent to combining per-horizon probabilities under an independence assumption ($1 - \prod_{h \leq H} (1 - q_h(x))$ overstates exposure because positions of the same member at successive horizons are strongly dependent). The distinction matters most exactly where humanitarian stakes are highest: near landfall, where along-track uncertainty spreads a single coherent threat across many horizons. Reporting both quantities separates *instantaneous horizon risk* (what might be affected at 48 h) from *cumulative planning risk* (what might be affected at any point through 48 h), and the evaluation of Section V reports both.

Exposure probabilities distinguish locations strongly supported by the ensemble (q near 1) from locations inside a broad uncertainty envelope but supported by few members (q near $1/M_h$). Binary-region approaches assign both the same status; Section V quantifies what this distinction is worth.

E. Probability-Weighted Exposure Estimation

Expected population exposure at horizon h is the probability-weighted sum over the analysis representation,

$$\widehat{\text{Pop}}(h) = \sum_{x \in \mathcal{X}} P(x) q_h(x), \quad (6)$$

which is exactly the ensemble mean of the per-member exposed populations and therefore inherits a direct interpretation: the expected number of people within the impact radius of the cyclone position at lead time h , under the ensemble's distribution over tracks. Vulnerability-weighted expected exposure replaces $P(x)$ with $P(x)V(x)$, and expected infrastructure exposure replaces it with the per-cell infrastructure count:

$$\widehat{\text{VW}}(h) = \sum_{x \in \mathcal{X}} P(x) V(x) q_h(x), \quad \widehat{\text{Infra}}(h) = \sum_{x \in \mathcal{X}} I(x) q_h(x). \quad (7)$$

The baseline estimators evaluated in Section IV instantiate the same functional with degenerate exposure fields. The *deterministic* estimator uses the mean track only,

$$q_h^{\text{det}}(x) = \mathbb{I}[d(x, \bar{\mathbf{p}}(h)) \leq r], \quad (8)$$

and the *P90-envelope* estimator treats all locations within the buffered ninetyth-percentile cone as exposed,

$$q_h^{\text{P90}}(x) = \mathbb{I}[d(x, \bar{\mathbf{p}}(h)) \leq r_{0.9}(h) + r]. \quad (9)$$

Framing all three approaches — mean track, P90 envelope, and member-level probability — as the same functional with different q makes the comparison in Section V an apples-to-apples measurement of what the probability weighting itself contributes.

F. Regional Aggregation and Prioritization

Humanitarian planning operates on administrative units, not grid cells. For each region R_j and horizon h , AOTS2Action computes the vulnerability-weighted expected exposure

$$S_j(h) = \sum_{x \in R_j} P(x) V(x) q_h(x), \quad (10)$$

together with a decomposition reported alongside every ranking: expected exposed population $\sum_{x \in R_j} P(x) q_h(x)$, regional mean exposure probability, and expected exposed infrastructure $\sum_{x \in R_j} I(x) q_h(x)$. Regions are ranked by vulnerability-weighted expected exposure to produce the priority list evaluated in Section IV. We deliberately avoid claiming that this scalar represents total humanitarian risk; it is a transparent, decomposable prioritization signal whose agreement with retrospectively realized priorities is measured rather than assumed. The scoring function is frozen before the retrospective evaluation is run.

Reporting the decomposition rather than the scalar alone serves the veracity requirement: a region may rank highly because many people face moderate probability or because few people face near-certain impact, and these call for different preparedness postures.

G. Implementation and Complexity

For each forecast horizon, AOTS2Action evaluates the great-circle distance between each ensemble-member position and every element of the analysis grid. For M_h members and $|\mathcal{X}|$ spatial elements, the probability-weighted exposure computation therefore requires $O(M_h |\mathcal{X}|)$ distance evaluations per horizon, and $O(\sum_h M_h |\mathcal{X}|)$ over a forecast cycle.

The implementation vectorizes distance computation across all spatial elements for each ensemble member and accumulates member-level exposure indicators into a single probability array. Consequently, the dominant working memory for exposure computation is $O(|\mathcal{X}|)$ rather than $O(M_h |\mathcal{X}|)$; individual member masks are not retained after accumulation.

The research implementation is written in Python using NumPy, pandas, SciPy, and scikit-learn. Experimental outputs, including case-level and aggregate calibration, exposure, ranking, and scalability results, are persisted as tabular artifacts with experiment settings specified through configuration files.

IV. RETROSPECTIVE EVALUATION METHODOLOGY

We evaluate AOTS2Action retrospectively around four research questions that test every link in the chain from forecast uncertainty to humanitarian priorities: **RQ1** — do the uncertainty representations calibrate (coverage and sharpness)? **RQ2** — does preserving ensemble uncertainty improve exposure estimates (absolute error, signed error, and Brier score against realized-track exposure)? **RQ3** — does this improve regional prioritization (nDCG, Recall, and Spearman correlation against realized rankings)? **RQ4** — does the computation scale (runtime, memory, and throughput in ensemble size and spatial volume)? The unifying principle is that every forecast-time estimate is compared against a quantity computed from the realized cyclone trajectory under the *same* operators the framework applies to forecasts, so that differences measure forecast quality and uncertainty handling rather than differences in definitions.

A. Data and Forecast–Observation Matching

The evaluation uses [ENSEMBLE SOURCE] ensemble forecasts for [N] tropical cyclones in the [BASIN(S)] basin(s)

during [YEARS], comprising [C] forecast cycles with up to [M] members each and horizons

$$\mathcal{H} = \{6, 12, 24, 48, 72, 96\} \text{ h.} \quad (11)$$

The short-range horizons (6–24 h) characterize near-term calibration, while 48–96 h expose how positional uncertainty grows into the preparedness-relevant range. Observed trajectories are taken from [IBTrACS / HURDAT2 / JTWC] best-track data.

For each forecast cycle initialized at t_0 and horizon $h \in \mathcal{H}$, the verifying position $\mathbf{p}^*(h)$ is obtained from the nearest available best-track record valid at $t_0 + h$, subject to a maximum temporal mismatch of 3 h. Forecast–horizon pairs without a verifying observation within this tolerance are excluded. The number of retained verifying pairs n_h and the number of distinct cyclones are reported separately at each horizon. When best-track records are six-hourly and forecast valid times fall on synoptic hours, most matches are exact; the tolerance governs only off-synoptic and end-of-lifetime cases.

B. RQ1: Calibration, Sharpness, and Track Error

For the percentile cone and covariance ellipse at nominal levels $\alpha \in \{0.5, 0.9\}$, empirical coverage at horizon h is the fraction of verifying pairs whose observed position falls inside the region:

$$\text{Cov}_\alpha(h) = \frac{1}{n_h} \sum_{i=1}^{n_h} \mathbb{1}[\mathbf{p}_i^*(h) \in A_{\alpha,i}(h)], \quad (12)$$

where i indexes verifying pairs and $A_{\alpha,i}(h)$ is the region constructed from cycle i 's ensemble. Absolute calibration error is $|\text{Cov}_\alpha(h) - \alpha|$. The fixed-radius baseline makes no nominal coverage claim and its region does not vary with α ; it therefore contributes a single empirical coverage and area per horizon, reported for reference.

Sharpness is the geographic area of the uncertainty region, $|A_\alpha(h)|$, averaged over cycles. Coverage and sharpness are reported jointly: high coverage can be obtained trivially by constructing excessively large regions, whereas arbitrarily small regions may be sharp but poorly calibrated.

Track error is the great-circle distance $d(\bar{\mathbf{p}}_i(h), \mathbf{p}_i^*(h))$ of the ensemble-mean track, reported as mean and median per horizon. This contextualizes the calibration results against operational verification practice and permits sanity comparison with published track-error climatologies for the same basins and seasons.

C. RQ2: Exposure-Estimation Fidelity

Ground truth for exposure is defined by applying the framework's own exposure operator to the realized trajectory: the realized exposure field treats the observed track as a degenerate single-member ensemble,

$$q_h^*(x) = \mathbb{1}[d(x, \mathbf{p}^*(h)) \leq r], \quad (13)$$

with realized population exposure $\text{Pop}^*(h) = \sum_x P(x) q_h^*(x)$ and realized vulnerability-weighted exposure $\text{VWExp}^*(h) = \sum_x P(x) V(x) q_h^*(x)$. Using the same operator and the same

impact radius r on both sides ensures that forecast–realized differences reflect track-uncertainty handling, not definitional mismatch.

Vulnerability-weighted exposure is the primary exposure quantity of this evaluation, as it most directly expresses the humanitarian objective; population exposure is reported as a secondary quantity under identical metrics. For each estimator $e \in \{\text{det}, \text{P90}, \text{ens}\}$ (Eqs. (8), (9), (4)) and each forecast cycle, we report absolute error, signed error, and exposure ratio at each horizon,

$$\text{AE}_e(h) = \left| \widehat{\text{VW}}_e(h) - \text{VWExp}^*(h) \right|, \quad \text{SE}_e(h) = \widehat{\text{VW}}_e(h) - \text{VWExp}^*(h) \quad (14)$$

with the ratio defined only for $\text{VWExp}^*(h) > 0$. Signed error reveals whether a method is systematically conservative or underestimates exposure; the ratio expresses the same tendency scale-free. Forecast cases with zero realized exposure are retained for absolute and signed error evaluation but excluded from exposure-ratio summaries; their number is reported separately at each horizon.

Beyond scalar totals, we evaluate the spatial quality of each estimator’s exposure field. Each estimator induces a field $q_h^e(x)$ — binary for the deterministic and P90 baselines ($q_h^{\text{det}}, q_h^{\text{P90}} \in \{0, 1\}$), graded for the ensemble ($q_h^{\text{ens}} \in [0, 1]$) — and all three are scored by the same Brier score against the realized indicators:

$$\text{BS}_e(h) = \frac{1}{|\mathcal{X}_{i,h}^{\text{eval}}|} \sum_{x \in \mathcal{X}_{i,h}^{\text{eval}}} (q_h^e(x) - q_h^*(x))^2, \quad (15)$$

with lower values better, so the comparison is fully symmetric and quantifies the value of graded ensemble support. To prevent the spatial score from being dominated by geographically irrelevant zero–zero cells, Brier scores are computed over a pre-specified storm-centered evaluation domain $\mathcal{X}_{i,h}^{\text{eval}}$ that is fixed independently of the estimator being evaluated. Together with Section IV-B, this yields a two-level uncertainty story: track-position calibration first, exposure probability calibration second — the first two links of the chain this evaluation is designed to test. Exposure reliability diagrams (binning cells by predicted q_h and plotting realized frequency) are reported as secondary analyses in the supplement.

D. RQ3: Regional-Ranking Quality

The retrospective target ranking for each forecast cycle orders regions by realized vulnerability-weighted exposure, $S_j^*(h) = \sum_{x \in R_j} P(x)V(x)q_h^*(x)$, i.e., Eq. (10) evaluated on the realized field. Predicted rankings order regions by $S_j(h)$ computed from each estimator’s exposure field.

For $\text{nDCG}@K$, the realized vulnerability-weighted exposure $S_j^*(h)$ is used directly as the nonnegative relevance gain; no post-hoc binning or thresholding of regional scores is applied. Recall@ K is the fraction of the top- K regions by realized score that appear in the predicted top- K . We report $K \in \{5, 10\}$ to evaluate both a narrow high-priority shortlist and a broader regional prioritization set, noting that

$J = [\text{number of regions}]$ administrative regions bound the meaningful range of K .

Spearman rank correlation is computed over the nonzero-union region set

$$\mathcal{R}_{i,h} = \{R_j : S_j(h) > 0 \vee S_j^*(h) > 0\}, \quad (16)$$

which prevents large numbers of zero–zero regions (inevitable at long lead times when the cyclone is far from land) from inflating apparent agreement. Spearman correlation is reported only when $|\mathcal{R}_{i,h}| \geq 3$; otherwise the case is marked undefined and excluded from correlation summaries, with the number of excluded cases reported alongside the correlation.

E. RQ4: Scalability

Scalability experiments measure wall-clock runtime, peak memory, and throughput ($M|\mathcal{X}|/T$ member–cell evaluations per second) for the exposure computation over a two-dimensional grid of ensemble size and spatial volume. With M_{full} the full ensemble size and $N_{\mathcal{X}}$ the full analysis-grid size,

$$M \in \{5, 10, 20, 40, M_{\text{full}}\}, \quad |\mathcal{X}| \in \{0.1, 0.25, 0.5, 0.75, 1.0\} \times N_{\mathcal{X}}, \quad (17)$$

where ensemble size is varied by subsampling members and spatial volume by **[uniform subsampling / cropping of the analysis grid]**. Each $(M, |\mathcal{X}|)$ setting is repeated 10 times; both median and mean runtime are reported, the median being robust to occasional system-induced timing spikes. We present runtime as a function of M at fixed $|\mathcal{X}|$, runtime as a function of $|\mathcal{X}|$ at fixed M , and throughput across the grid, verifying empirically the $O(M_h|\mathcal{X}|)$ scaling predicted in Section III-G; the practical criterion is that a full ensemble–geospatial integration completes well within one forecast cycle.

Experiments were executed on a workstation with **[CPU]**, **[RAM]** GB memory, running **[OS]**. The core exposure computation is CPU-based and uses NumPy vectorized distance evaluation.

F. Statistical Protocol and Pre-Specified Evaluation Conditions

a) *Inference unit.*: Forecast cycles within one cyclone share the realized trajectory and are strongly dependent, so the primary inference unit is the *cyclone*. For a paired comparison between estimators a and b at horizon h , per-cycle differences are first averaged within each storm s ,

$$\Delta_{s,h} = \frac{1}{C_{s,h}} \sum_{i \in \text{cycles}(s)} (\varepsilon_{a,i,h} - \varepsilon_{b,i,h}), \quad (18)$$

and the Wilcoxon signed-rank test is applied to the storm-level differences $\{\Delta_{s,h}\}_{s=1}^N$. Confidence intervals (95%) for all aggregate metrics are obtained by a storm-level bootstrap with $B = 10,000$ replicates: cyclones are resampled with replacement, a resampled cyclone contributes *all* of its forecast cycles, a cyclone may appear multiple times within one replicate, and the aggregate metric is recomputed per replicate.

b) *Multiplicity.*: Holm step-down correction at family-wise level 0.05 is applied within pre-defined families. The primary exposure family comprises the two comparisons (det – ens) and (P90 – ens) on vulnerability-weighted absolute error at the three primary horizons (six tests). Ranking comparisons form a separate family. Secondary metrics and horizons are reported with unadjusted p -values and clearly labeled as secondary; no correction family aggregates unrelated hypotheses. Adjusted p -values are reported with full precision, and all per-horizon results appear in the supplement regardless of significance.

c) *Pre-specified conditions.*: The following evaluation protocol was fixed, and committed to the project repository, before the final evaluation was executed:

- *RQ1 (calibration).* The primary calibration endpoint is P90 empirical coverage and region area of the **[REPRESENTATION — O2, fix a priori]** at 48 h, with full coverage–sharpness curves across all six horizons reported for all representations. The abstract’s joint coverage-and-sharpness sentence is used only if the primary P90 coverage is within $\pm$ **[TOL — O1, freeze after sample-size audit]** of nominal; otherwise the pre-written fallback calibration-tradeoff sentence is used.
- *RQ2 (exposure).* The primary exposure comparison evaluates $AE_{\text{det}}^{\text{VW}} - AE_{\text{ens}}^{\text{VW}}$ at 24, 48, and 72 h, with $AE_{\text{P90}}^{\text{VW}} - AE_{\text{ens}}^{\text{VW}}$ as the second family member. The short-range horizons (6–12 h) are near-term and comparatively easy; 96 h is a valuable secondary long-range analysis.
- *RQ3 (ranking).* The pre-specified headline regional-ranking metric is nDCG@10 at 48 h — long enough for uncertainty propagation to matter, while operationally meaningful for anticipatory action.
- *RQ4 (scalability).* Runtime and throughput over the $M \times |\mathcal{X}|$ grid with 10 repetitions per setting.
- *Secondary outcomes.* Population exposure, signed error, exposure ratio, Brier score, nDCG@5, Recall@5, Recall@10, Spearman correlation, and all analyses at 6, 12, and 96 h.
- *Null results.* If the ensemble estimator does not improve on a baseline for any pre-specified primary comparison, the result is reported as such in the main text.

V. RESULTS

A. Dataset and Verification Sample

Table I summarizes the evaluation corpus: the storms, forecast cycles, and, at each horizon, the number of verifying (cycle, horizon) pairs n_h and distinct cyclones after applying the 3 h matching tolerance of Section IV-A. Attrition at long horizons reflects storm dissolution and end-of-lifetime censoring rather than data quality issues.

B. RQ1: Uncertainty Calibration and Sharpness

Table ?? reports empirical coverage and mean region area at 48 and 96 h for all representations; Figure ?? shows the

TABLE I
EVALUATION CORPUS AND PER-HORIZON VERIFICATION SAMPLE.
VERIFYING PAIRS ARE (CYCLE, HORIZON) COMBINATIONS WITH A
BEST-TRACK OBSERVATION WITHIN THE 3 H MATCHING TOLERANCE.

	6 h	12 h	24 h	48 h	72 h	96 h
Verifying pairs n_h						
Distinct cyclones						
Zero-realized-exposure cases						
Cases with $ \mathcal{R}_{i,h} < 3$						

TABLE II
VULNERABILITY-WEIGHTED EXPOSURE ABSOLUTE ERROR (MEAN [95%
CI] OVER STORMS) BY ESTIMATOR AND LEAD TIME. PAIRED ROWS GIVE
STORM-LEVEL DIFFERENCES FOR THE PRE-SPECIFIED SIX-TEST FAMILY
(HOLM-ADJUSTED p). POPULATION-EXPOSURE AND
REMAINING-HORIZON RESULTS APPEAR IN THE SUPPLEMENT.

	24 h	48 h	72 h
<i>Absolute error, AE^{VW}</i>			
Deterministic (mean track)			
P90 envelope			
Ensemble prob.-weighted			
<i>Paired differences (Holm-adjusted p)</i>			
det – ens			
P90 – ens			

full coverage and sharpness curves across horizons, with the six-horizon table in the supplement.

The primary calibration endpoint is P90 coverage and area of the **[REPRESENTATION (O2)]** at 48 h: coverage **[X]** (95% CI **[lo, hi]**) with mean region area **[A]** km². The representation is empirically well calibrated at the primary endpoint, and attains this coverage with **[Y%]** smaller regions than **[comparison representation / fixed radius]**, i.e., comparable reliability at materially higher sharpness. At the primary endpoint the representation attains **[X]** empirical P90 coverage with **[Y%]** smaller regions than **[comparison]**, quantifying a calibration–sharpness tradeoff: **[one sentence stating the direction of miscoverage and its horizon dependence, drafted from the curves — both under- and over-coverage variants below]**.

Track error of the ensemble-mean track (Table ??, supplement if space requires) grows from **[.]** km at 6 h to **[.]** km at 96 h, consistent in order of magnitude with published operational verification for **[basin, period]**, indicating that the evaluation corpus is not anomalous.

C. RQ2: Exposure-Estimation Fidelity

Table II reports the primary comparison: vulnerability-weighted absolute exposure error by estimator and horizon, with the paired storm-level differences and Holm-adjusted p -values for the pre-specified family.

The ensemble estimator reduces vulnerability-weighted exposure error relative to the deterministic mean-track baseline by **[Z%]** at **[h]** h (Holm-adjusted $p = \text{[.]}$), with **[consistent / attenuating / growing]** differences across the primary horizons. Against the P90 envelope, probability weighting reduces

TABLE III
BRIER SCORE OF THE EXPOSURE FIELD AGAINST REALIZED EXPOSURE (LOWER IS BETTER), COMPUTED OVER THE PRE-SPECIFIED STORM-CENTERED EVALUATION DOMAIN, BY ESTIMATOR AND LEAD TIME.

	6 h	12 h	24 h	48 h	72 h	96 h
Deterministic (mean track)						
P90 envelope						
Ensemble prob.-weighted						

TABLE IV
REGIONAL-RANKING AGREEMENT WITH REALIZED VULNERABILITY-WEIGHTED PRIORITIES (MEAN [95% CI] OVER STORMS). THE PRE-SPECIFIED HEADLINE IS NDCG@10 AT 48 h (BOLD COLUMN). SPEARMAN IS COMPUTED OVER THE NONZERO-UNION REGION SET WITH $|\mathcal{R}_{i,h}| \geq 3$.

Metric	Estimator	24 h	48 h	72 h
nDCG@10	Det. / P90 / Ens.			
nDCG@5	Det. / P90 / Ens.			
Recall@10	Det. / P90 / Ens.			
Recall@5	Det. / P90 / Ens.			
Spearman	Det. / P90 / Ens.			

error by [..] at [h] h, reflecting the envelope’s systematic over-estimation (below). At the pre-specified primary horizons, the ensemble estimator does not significantly reduce vulnerability-weighted exposure error relative to **[baseline]** (Holm-adjusted $p = [..]$); we report this null result directly and note that **[secondary observation drafted from the signed-error / Brier structure, if the graded field shows value the scalar totals do not]**.

Figure ?? shows the systematic behavior behind the absolute errors: signed error and exposure ratio by estimator and horizon. The P90 envelope **[overestimates realized exposure by a factor of [..] at 72 h / behavior as observed]**, while the deterministic track **[under/over-estimates with high dispersion / as observed]**; the ensemble estimator’s ratio remains **[nearest unity across horizons / as observed]**.

Table III completes the two-level calibration story: Brier scores over the storm-centered evaluation domain for all three estimators. The graded ensemble field achieves the lowest Brier score at **[k of 6]** horizons, quantifying the value of expressing differential forecast support that binary fields cannot represent; reliability diagrams in the supplement show the ensemble probabilities to be **[well calibrated / over- or under-confident as observed]**.

D. RQ3: Regional Prioritization

Table IV reports ranking agreement against realized regional priorities.

At the pre-specified headline endpoint, the ensemble estimator attains $\text{nDCG@10} = [\mathbf{R}]$ (95% CI **[lo, hi]**) at 48 h, exceeding the deterministic and P90 baselines by [..] and [..] respectively (Holm-adjusted $p = [..]$ within the ranking family); the advantage **[grows with horizon / pattern as observed]**, consistent with uncertainty propagation mattering

most where positional spread is largest. Recall@10 shows that [..] of the ten highest-priority regions by realized exposure are identified in advance at 48 h.

E. RQ4: Scalability

Figure ?? reports runtime and throughput over the $M \times |\mathcal{X}|$ grid (10 repetitions per setting; median shown, mean in the supplement). Runtime scales **[linearly, slope [..] on log-log axes]** in both ensemble size and spatial volume, matching the $O(M_h|\mathcal{X}|)$ analysis of Section III-G; peak memory remains **[flat in M at [..] GB]**, confirming the $O(|\mathcal{X}|)$ accumulation design. A full integration at $(M_{\text{full}}, N_{\mathcal{X}})$ completes in **[T] [s/min]** on the workstation of Section IV-E — **[well within]** one forecast cycle, the practical criterion for operational use.

VI. DISCUSSION

AOTS2Action was designed around a simple question: what is gained by preserving ensemble uncertainty beyond the forecast itself, rather than reducing it to a representative track or uncertainty envelope before downstream humanitarian analysis? The retrospective evaluation tests this question at each link of the chain from forecast uncertainty to calibrated spatial probabilities to exposure estimates to regional priorities, and the interpretation below follows the pre-specified hierarchy of Section IV-F: primary endpoints first, secondary outcomes as context, and null results reported at the same level of prominence as positive ones. The results show **[overall summary: e.g., that the value of retaining member-level uncertainty is measurable downstream / a mixed pattern whose benefit depends on horizon and evaluation target]**.

A. From Track Calibration to Exposure Calibration

The RQ1 results highlight that uncertainty representations should not be evaluated by coverage alone. A sufficiently large region can obtain high empirical coverage while providing little geographic discrimination, whereas a compact representation may appear sharp but systematically exclude the realized cyclone position. The joint coverage–sharpness evaluation therefore provides a more meaningful comparison of the percentile cone and covariance ellipse.

At the primary 48 h endpoint, the **[REPRESENTATION]** achieved the strongest calibration–sharpness tradeoff, with empirical P90 coverage of **[X]** and a mean area of **[A] km²**. Relative to **[comparison]**, this corresponds to **[Y% smaller/larger]** regions at **[similar/higher/lower]** empirical coverage. This result suggests that **[e.g., explicitly representing anisotropic ensemble dispersion preserves useful directional structure that a circular envelope cannot capture]**.

Neither uncertainty representation uniformly dominates: the **[METHOD A]** provides **[better coverage]**, whereas the **[METHOD B]** is **[sharper]**. This tradeoff is itself operationally relevant because uncertainty regions are not useful solely as statistical objects; excessively broad regions increase the number of communities that must be treated as plausibly affected.

More importantly, RQ2 shows whether calibration at the track level survives the transformation into humanitarian exposure. This is a different question from conventional trajectory verification. A forecast can have reasonable position coverage while still yielding poor exposure estimates when small spatial deviations intersect strongly heterogeneous population or vulnerability fields. Conversely, modest position errors may have little humanitarian effect when plausible tracks traverse sparsely populated areas. The exposure-field Brier score therefore complements track coverage by evaluating uncertainty in the space in which downstream decisions are made.

The member-level ensemble field achieves lower Brier error than the binary alternatives at **[K of 6]** horizons, indicating that graded ensemble support contains useful information beyond membership in an uncertainty envelope. In particular, a location supported by **[e.g., 80%]** of members and one supported by **[e.g., 5%]** of members may both fall inside a P90 envelope, but the two should not receive the same exposure probability.

The ensemble field does not uniformly improve Brier score across horizons. At **[short horizons]**, the ensemble is sufficiently concentrated that a binary representation **[matches/outperforms]** graded probabilities, while the advantage of member-level support emerges primarily at **[longer horizons / particular cases]**. This limits the claim to the forecast regimes in which ensemble dispersion contains decision-relevant spatial information.

B. Why Probability Weighting Changes Exposure Estimates

The deterministic, P90-envelope, and ensemble-probability estimators represent three distinct assumptions about forecast uncertainty. The deterministic estimator assumes that the ensemble mean is the realized trajectory; the P90 estimator treats every location inside a broad uncertainty envelope as equally exposed; and AOTS2Action retains the fraction of members supporting exposure at each location. Their differences are therefore not merely implementation choices but different probabilistic assumptions about the future.

The signed-error and exposure-ratio results clarify these assumptions. The P90 estimator is systematically conservative, **[overestimating realized vulnerability-weighted exposure by ... at ...]**, because it assigns unit exposure to all locations inside the envelope. The deterministic estimator exhibits **[underestimation / high variance / mixed bias]** because a single track cannot represent spatially separated plausible futures. The ensemble estimator remains **[closer to zero signed error / closer to an exposure ratio of one]** across the primary horizons, indicating that probability weighting provides a better compromise between these two extremes.

The observed bias pattern is more nuanced than the expected deterministic-versus-conservative contrast. **[Describe frozen result.]** This indicates that the value of probability weighting depends not only on ensemble dispersion but also on where that dispersion lies relative to heterogeneous population and vulnerability fields.

The distinction is particularly important for vulnerability-weighted exposure. Two forecast realizations with similar exposed populations can imply substantially different humanitarian priorities if those populations differ in vulnerability. By carrying uncertainty through $P(x)V(x)$ rather than adding vulnerability only after a deterministic exposure decision, AOTS2Action preserves this interaction at the spatial level.

At the same time, vulnerability-weighted exposure should not be interpreted as a complete measure of humanitarian loss. It is an expected exposure statistic conditioned on the forecast representation and the selected vulnerability layer. It does not directly estimate mortality, economic damage, displacement, service disruption, or household-level coping capacity. We therefore use it as a prioritization signal rather than a prediction of realized disaster severity.

C. From Exposure Accuracy to Regional Prioritization

A central motivation for AOTS2Action is that improved spatial probability estimates are useful only if they affect decisions at the geographic scales used for preparedness. RQ3 therefore evaluates whether exposure differences translate into better rankings of administrative regions rather than stopping at cell-level accuracy.

The improvement in nDCG@10 from **[DET/P90 value]** to **[ENS value]** at the pre-specified 48 h endpoint shows that the ensemble representation not only improves a continuous exposure metric but places more of the realized high-priority regions near the top of the forecast ranking. This is especially important because humanitarian planning typically operates under resource constraints: the practical question is not whether every region is ordered perfectly, but whether the highest-impact regions are identified early enough to receive attention.

Despite differences in exposure error, the ensemble estimator does not significantly improve the pre-specified nDCG@10 endpoint. This result indicates that better continuous exposure estimates do not automatically produce different top-ranked administrative regions. Regional aggregation may absorb cell-level differences when several methods assign similar total exposure to the same high-population areas. The null ranking result therefore places an important boundary on the practical value of uncertainty propagation.

The use of multiple ranking metrics also exposes different failure modes. nDCG emphasizes the ordering and magnitude of the highest-risk regions, Recall@ K asks whether the correct regions appear in a finite priority set, and Spearman correlation measures broader ordering over regions receiving nonzero forecast or realized score. Agreement in one metric does not imply agreement in the others. For example, **[insert result-specific interpretation if nDCG improves but Spearman does not, or vice versa]**.

D. Forecast Horizon and the Value of Ensemble Information

The value of ensemble uncertainty is expected to vary with lead time. At very short horizons, members may be tightly concentrated and the ensemble mean already provides a strong

approximation to the plausible track distribution. As horizon increases, track spread grows and the cost of collapsing the ensemble can increase. However, greater spread also makes probabilistic calibration harder.

The results exhibit **[describe frozen horizon pattern]**. The ensemble advantage increases from **[short]** to **[48/72]** h, supporting the hypothesis that member-level probabilities become most valuable once alternative trajectories are sufficiently separated to imply different exposure patterns. The benefit peaks near **[h]** h and declines at 96 h. This suggests a useful operating regime in which the ensemble contains meaningful geographic discrimination without being so diffuse or miscalibrated that probability mass becomes unreliable. No monotone horizon effect is observed, suggesting that geographic context—not horizon alone—determines whether ensemble spread changes humanitarian estimates.

This horizon dependence argues against treating “uncertainty-aware” as a binary property of a forecasting system. The relevant question is whether the uncertainty representation remains calibrated and decision-informative at the horizon at which preparedness actions are taken.

E. Scalability and the Big-Data Setting

AOTS2Action operates at the intersection of ensemble forecasting and large geospatial data, where computational demand grows with both the number of forecast members and the size of the spatial analysis domain. The empirical scalability results provide a practical test of the $O(M|\mathcal{X}|)$ exposure computation analyzed in Section III-G.

Runtime grows approximately linearly with both ensemble size and spatial volume, while peak memory remains **[approximately independent of M / describe result]**. This behavior is consistent with the implementation’s member-wise accumulation strategy, which avoids materializing a full $M \times |\mathcal{X}|$ tensor. The observed scaling departs from the ideal linear model at **[setting]**, where **[memory/cache/I/O/other]** becomes dominant. This identifies the principal engineering bottleneck for larger deployments.

At the largest evaluated configuration, the pipeline requires **[T]** to process **[M]** ensemble members over **[N]** spatial elements. Relative to the forecast-update interval of **[cycle duration]**, this provides **[carefully stated operational interpretation]**.

The scalability experiment is intentionally focused on the core member–geospatial computation rather than the interactive visualization layer. This isolates the cost introduced by preserving ensemble support, which is the incremental computation at the center of the methodological contribution.

F. Relationship to Existing Forecast and Impact Systems

AOTS2Action is not a new tropical cyclone forecasting model. It is forecast-source-agnostic: operational numerical ensembles or learned ensemble systems can provide the upstream trajectories. Its contribution begins after those forecasts are produced, at the point where probabilistic trajectory information must be integrated with population, vulnerability, infrastructure, and administrative data.

This positioning distinguishes AOTS2Action from work focused on improving track forecasts or constructing better uncertainty regions. Probability ellipses, conformal regions, and probabilistic neural track models improve the representation of where a cyclone may travel; AOTS2Action asks how alternative uncertainty representations affect what is inferred downstream about exposed populations and geographic priorities.

It also differs from impact systems that begin from a deterministic hazard scenario or thresholded alert. The framework retains member-level support through exposure estimation, allowing the same location to receive different exposure probabilities depending on how many plausible trajectories support it. The evaluation then tests whether that additional information survives aggregation into decision-relevant regional outputs rather than assuming that it does.

G. Limitations

Several limitations constrain the interpretation of the results.

a) *Track-based hazard approximation.*: The current exposure operator represents potential impact through a distance threshold around the forecast cyclone position. This is a deliberate abstraction for evaluating uncertainty propagation, but a real cyclone footprint depends on intensity, storm size, asymmetric wind structure, rainfall, storm surge, terrain, and local infrastructure. Two trajectories passing equally close to a location may therefore produce substantially different impacts. Future work should replace the fixed-radius operator with ensemble-specific wind, rainfall, surge, or multi-hazard footprints where those variables are available.

b) *Realized exposure is a verification proxy.*: The retrospective target uses the observed cyclone trajectory with the same exposure operator used for forecasts. This construction provides a controlled comparison of uncertainty-handling methods, but it is not observed humanitarian impact ground truth. Realized population displacement, infrastructure disruption, casualties, and service failures would require event-specific impact datasets that are not uniformly available across the evaluation corpus.

c) *Population and vulnerability uncertainty.*: Population rasters and vulnerability indices are themselves estimates with different spatial resolutions, update schedules, and uncertainty. AOTS2Action currently treats these layers as fixed during each retrospective evaluation. Consequently, reported uncertainty reflects the cyclone ensemble but does not yet propagate uncertainty from the humanitarian data layers themselves.

d) *Administrative aggregation.*: Regional rankings depend on the administrative partition used for aggregation. Large regions may contain substantial internal heterogeneity, while small boundary changes can alter exposure totals near region edges. Rankings should therefore be interpreted at the resolution of the underlying administrative and population data rather than as invariant geographic truths.

e) *Ensemble calibration.*: Probability-weighted exposure inherits the calibration properties of the upstream ensemble. Member frequency is meaningful as a probability only to

the extent that the ensemble is appropriately calibrated and members represent plausible forecast uncertainty. The RQ1 and Brier evaluations diagnose this issue empirically but do not correct a miscalibrated upstream ensemble.

f) *Evaluation scope.*: The retrospective corpus contains [N] cyclones from [basin/year range]. Conclusions therefore apply to the evaluated forecast source, seasons, geographic coverage, and humanitarian datasets. Generalization across basins, forecasting systems, and vulnerability contexts requires additional evaluation.

g) *Computational scope.*: The current implementation uses vectorized great-circle calculations over a harmonized spatial representation. More sophisticated hazard footprints and much higher-resolution global rasters may require spatial indexing, tiled processing, distributed execution, or GPU acceleration beyond the implementation evaluated here.

H. Responsible Use

AOTS2Action is intended to support analysis, not to automate humanitarian decisions. Regional priority scores summarize forecast uncertainty, exposure, and vulnerability under explicit assumptions; they should not be interpreted as directives for resource allocation or evacuation. Operational decisions must also account for local knowledge, forecast-center guidance, infrastructure status, accessibility, institutional capacity, and information unavailable to the model.

The framework therefore exposes intermediate quantities—ensemble support, expected population exposure, vulnerability-weighted exposure, infrastructure exposure, and uncertainty representations—rather than presenting only a single opaque priority score. Dataset provenance and configuration information are retained with derived outputs so that users can trace how rankings were produced and identify the assumptions underlying them.

Finally, uncertainty should be communicated explicitly. A low-ranked region is not necessarily safe, particularly at long forecast horizons or under a poorly calibrated ensemble. The framework is designed to make uncertainty visible throughout the analytics chain rather than convert probabilistic forecasts into unwarranted certainty.

I. Discussion Summary

Taken together, the results show **[support / partial support / limited support]** for the central hypothesis that preserving ensemble uncertainty beyond the forecasting stage improves downstream humanitarian analysis. The strongest evidence is **[headline RQ1–RQ3 result]**, while **[important null result or limitation]** defines the boundary of the contribution.

VII. CONCLUSION

We presented **AOTS2Action**, an uncertainty-preserving geospatial analytics framework that connects tropical cyclone ensemble forecasts with downstream humanitarian exposure estimation and regional prioritization. Rather than collapsing an ensemble to a single representative trajectory or treating all locations within an uncertainty envelope as equally exposed,

AOTS2Action retains member-level forecast support and propagates it through population, vulnerability, infrastructure, and administrative data. The framework therefore provides a common pipeline for evaluating how alternative representations of track uncertainty affect not only positional calibration, but also the humanitarian quantities derived from them.

Our retrospective evaluation examines this propagation at four levels: uncertainty calibration and sharpness, exposure-estimation fidelity, regional-ranking quality, and computational scalability. At the pre-specified 48 h calibration endpoint, **[REPRESENTATION]** achieved P90 empirical coverage of **[COVERAGE]** with mean uncertainty-region area **[AREA]** km². For the primary exposure analysis, member-level probability weighting **[reduced / did not significantly reduce]** vulnerability-weighted exposure error relative to the deterministic mean-track baseline by **[X%]** at **[H]** h **[and corresponding results at the other primary horizons]**. Regional prioritization achieved nDCG@10 of **[R]** at the pre-specified 48 h endpoint, compared with **[BASELINE VALUES]**, while scalability experiments showed **[approximately linear / observed]** growth with ensemble size and spatial data volume. Together, these results **[support / partially support / place limits on]** the hypothesis that preserving forecast uncertainty beyond the forecasting stage improves downstream humanitarian analysis.

More broadly, the study demonstrates that uncertainty should be evaluated not only at the forecast trajectory itself, but also after it has been transformed into exposure estimates and decision-oriented regional summaries. A well-calibrated cyclone uncertainty region does not necessarily produce well-calibrated humanitarian exposure, and improved spatial probabilities do not automatically guarantee better regional rankings. Evaluating the complete chain

forecast uncertainty → spatial exposure probability
→ humanitarian exposure → regional prioritization

therefore provides a more direct assessment of whether probabilistic forecast information remains useful for downstream analysis.

AOTS2Action is complementary to, rather than a replacement for, operational or learned cyclone forecasting systems. Its contribution begins after an ensemble forecast is produced and focuses on preserving, integrating, evaluating, and communicating that uncertainty through heterogeneous geospatial data. Future work will extend the current track-based exposure operator to intensity- and structure-dependent hazard footprints, including wind, rainfall, and storm surge; propagate uncertainty from population and vulnerability layers in addition to the forecast ensemble; and evaluate the framework across additional basins, forecast systems, and event-level impact datasets. These extensions would move the framework from uncertainty-aware exposure analysis toward more comprehensive probabilistic impact forecasting while retaining the transparent evaluation principles developed here.

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